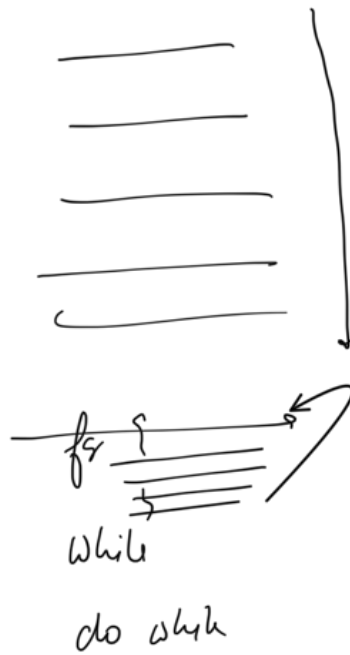
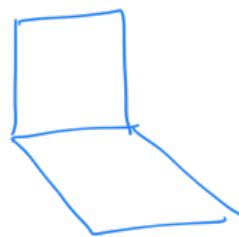
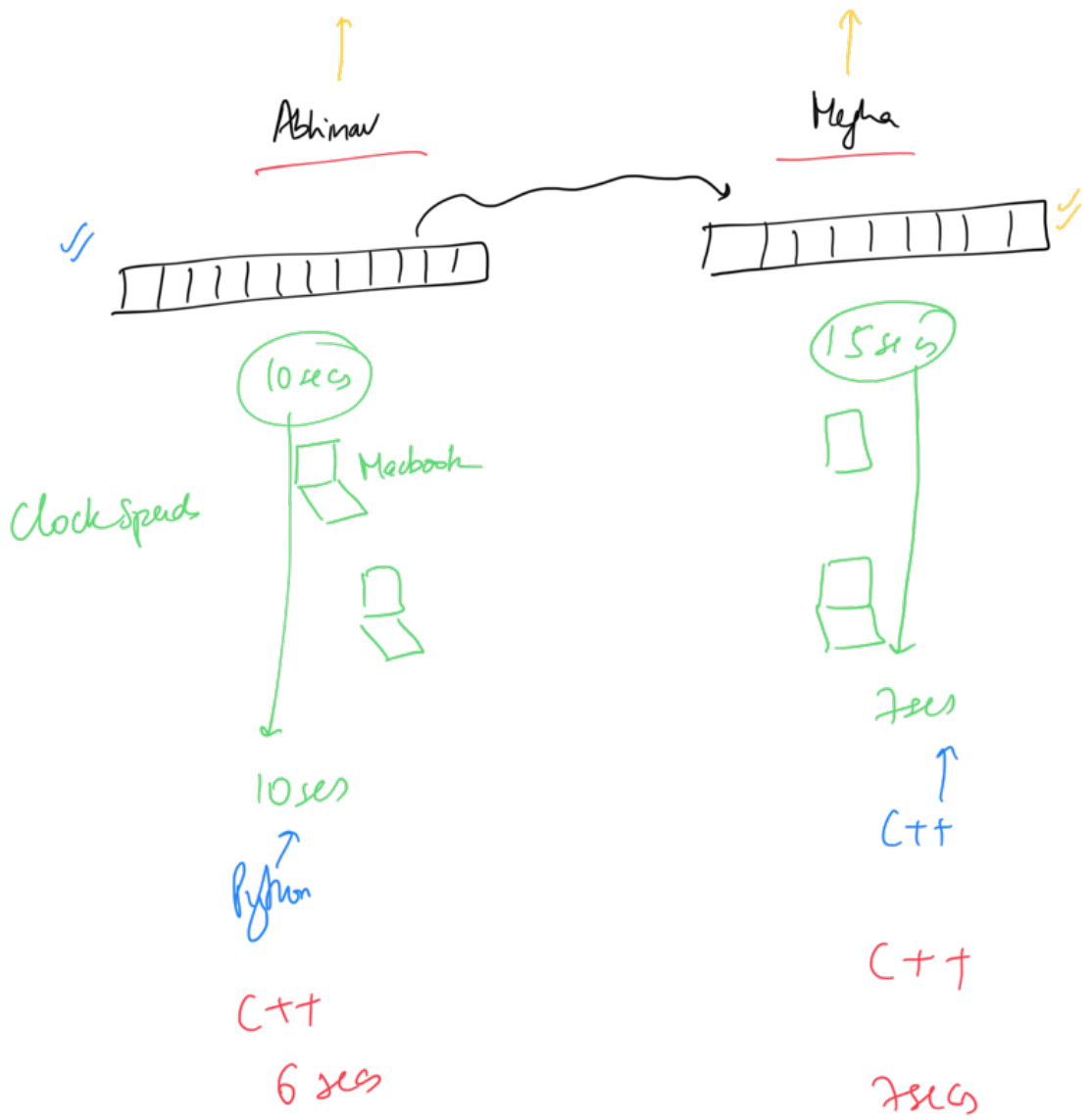


Time Complexity - 2

Addition	$a++$
Assignment	$a = 7$
Comparison	$a == 7$

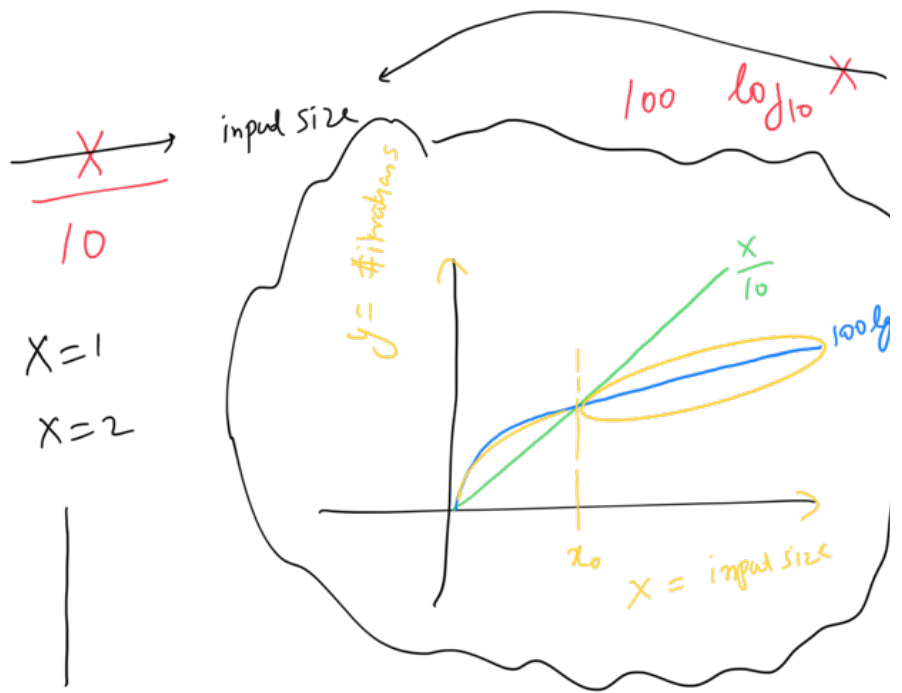
TLE ☹️ ☹️





$(\cdot \sim)$

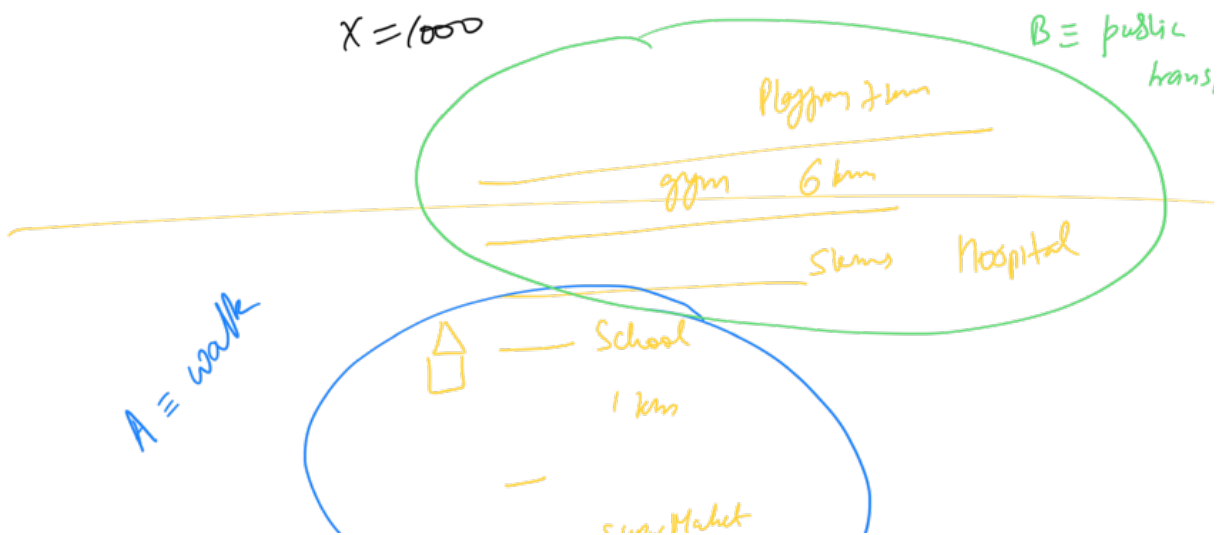
$(\cdot \propto)$



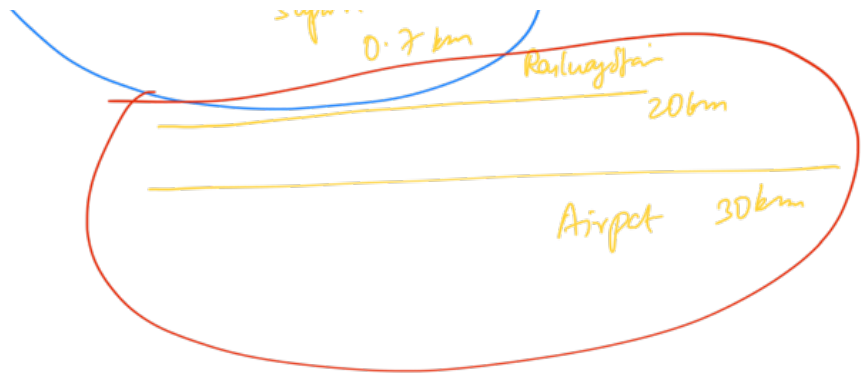
$X = 10$

$X = 100$

$X = 1000$



Driving $\equiv C$
Gas



iterations

$N \equiv$ size input

II

growth = linear

(A)	N
(B)	$10 + N$
(C)	$100 + N$
(D)	$10N$

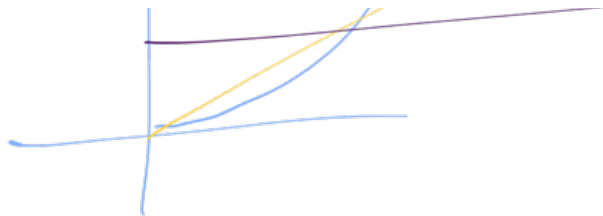
growth = 0

I

(E)	70
(F)	700

III

(G)	$10 \times N^2$
(H)	$5 \times N^2$



$$\frac{x}{10}$$

$$100 \log_{10} x$$

$$x = 10$$

$$\frac{10}{10} = 1$$

$$10^0$$

(move him)

can
him

$$x = 100$$

$$\frac{100}{10} = 10$$

$$100 \times \log_{10} 100$$

$$= 200$$

$$x = 1000$$

$$\frac{1000}{10} = 100$$

$$100 \times \log_{10} 1000$$

$$= 300$$

$$X = 10000$$

$$\frac{10k}{10} = 1000$$

$$100 \times \log_{10} 10000$$

$$= 400$$

$$X = 100000$$

$$10000$$

$$100 \times \log_{10} 100000$$

$$= 500$$

100k

600

100k

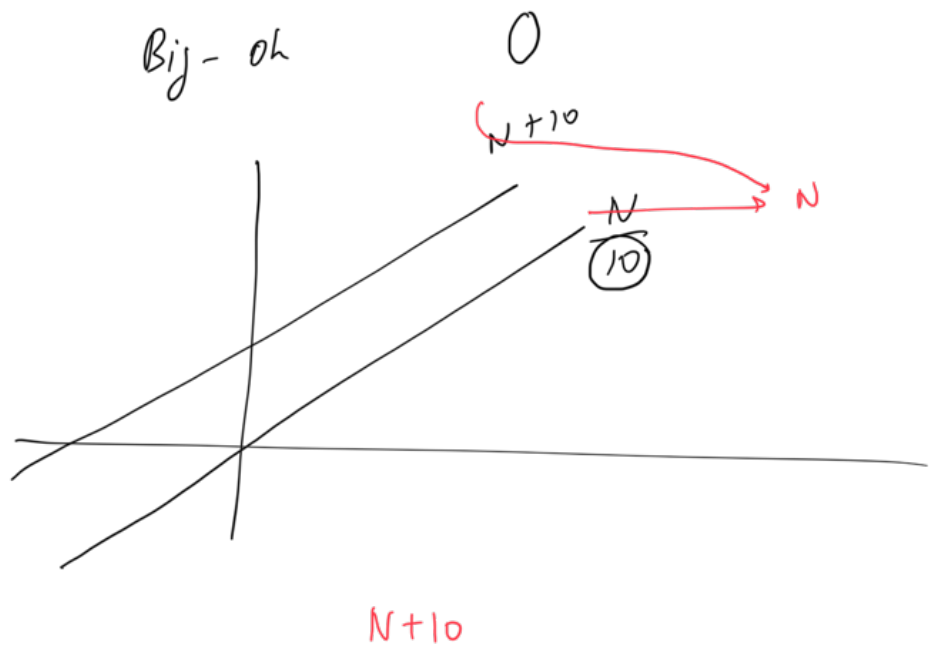
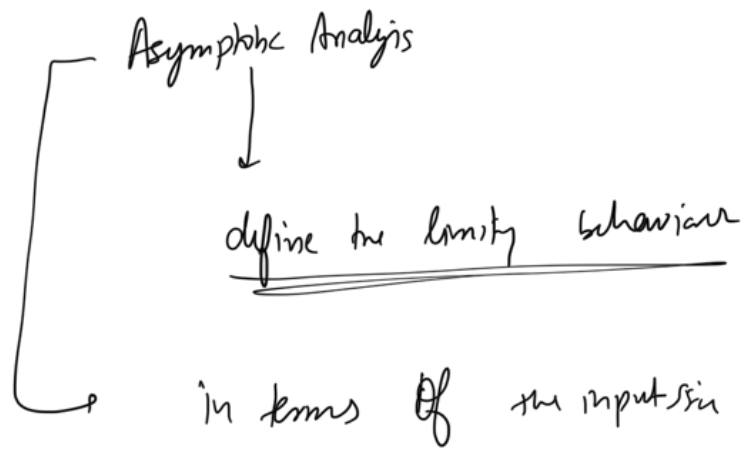
700

① Growth

wrt your input growth

② Buckets

Compare Buckets



$$\textcircled{N} = 10000$$

10010

$$\textcircled{N} = 100k$$

100k+10

$$\begin{array}{l} (N+1)M \\ (M+1) \end{array}$$

$$\begin{array}{l} 10 \times N \\ N = 10k \\ 100k \end{array} = N$$

$$\begin{array}{l} N = 100k \\ 1M \end{array}$$

$$\begin{array}{l} N = 1M \\ 10M \end{array}$$

$$\begin{array}{r} \frac{\frac{10N}{10+N}}{100N} \\ \hline \frac{1000+N}{10+10N} \end{array} = \underline{\underline{O(N)}}$$

$$\begin{array}{l} 10N^2 \\ 50N^2 \\ \dots 2 \end{array} = \underline{\underline{O(N^2)}}$$

$$\begin{array}{c}
 1000N \\
 1000 + N^2 \\
 10 + 100N^2
 \end{array}$$

$$\begin{array}{c}
 N^3 \\
 10N^3
 \end{array}$$

$$= \underline{\underline{O(N^3)}}$$

$$\begin{array}{r}
 5000 \\
 \hline
 70k \\
 \hline
 10
 \end{array}$$



$$\underline{\underline{O(1)}}$$

$$\begin{array}{l}
 N=10 \\
 N=1000 \\
 N=10k
 \end{array}$$



Big - oh Notation

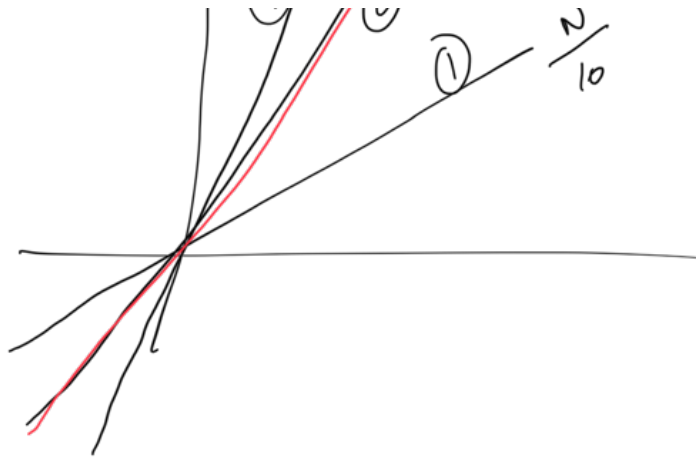
O

formal way to express the upper bound on algo running time.

Tells us the worst case time complexity of my algo

$$\left(\begin{array}{c} 10N \\ N \end{array} \right) \quad \text{with a red diagonal line through the } N$$

$$= O(N)$$



✓ $\boxed{\frac{N}{10}} = O(N)$ → 100k N

✓ $\boxed{\frac{N}{100}} \rightarrow O(N)$

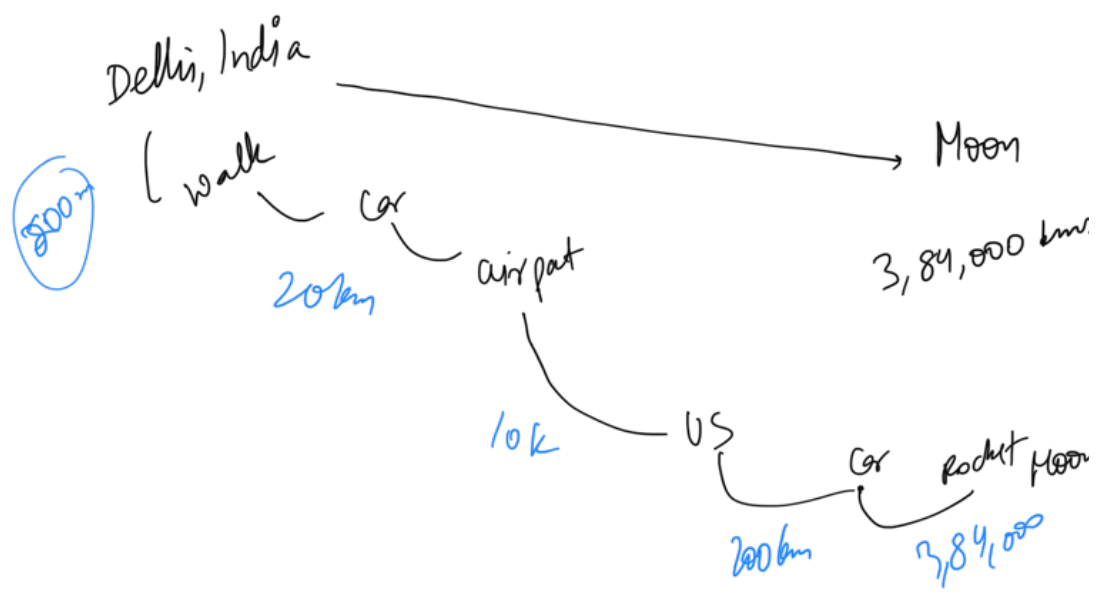
✓ $\textcircled{10N} \rightarrow O(N)$

$O(\underline{f(n)}) = \{ g(n) : \text{there exist } \underline{c} > 0$
 $\underline{f(n)} \leq \underline{c} \times \underline{g(n)}$
 $\text{for } \underline{x > x_0}$

$\textcircled{10N} \rightarrow O(N)$
 $\textcircled{10k \times N}$

~~$O(N)$~~

look N
 $O(N)$



3,84,000
+ n
+ n
+ n
+ n

$\sim 3,84k$

$O(N) =$

$N + 10$

$O(N)$ $O(1)$

$$O(N^2) = 1000 N^2 + 10N + 5$$

Diagram showing the asymptotic complexity of the expression $1000 N^2 + 10N + 5$. Arrows point from each term to its complexity: $1000 N^2 \rightarrow O(N^2)$, $10N \rightarrow O(N)$, and $5 \rightarrow O(1)$.

$$10^6 = O(1)$$

$$f(n) = 10^6$$

$$O(f(n)) = O(1)$$

$$4N^2 + 3N + 10 \neq O(N^2)$$

Diagram showing the asymptotic complexity of the expression $4N^2 + 3N + 10$. Arrows point from each term to its complexity: $4N^2 \rightarrow O(N^2)$, $3N \rightarrow O(N)$, and $10 \rightarrow O(1)$. The entire expression is boxed and labeled $\neq O(N^2)$.

$$N^2 + 10N = O(N^2)$$

Diagram showing the asymptotic complexity of the expression $N^2 + 10N$. Arrows point from each term to its complexity: $N^2 \rightarrow O(N^2)$ and $10N \rightarrow O(N)$. The entire expression is labeled $= O(N^2)$.

✓ Constant = $O(1)$
 log $O(\log_2 N)$



✓ linear = $O(N)$
 $O(N \log N)$



quadratic = $O(N^2)$
 cubic = $O(N^3)$
 polynomial = $O(N^4)$
 $O(N^{10})$



Exponential = 2^n
 4^n
 6^n

$$10^4$$

$$11^4$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

$$2^6 = 64$$

$$2^7 = 128$$

$$2^8 = 256$$

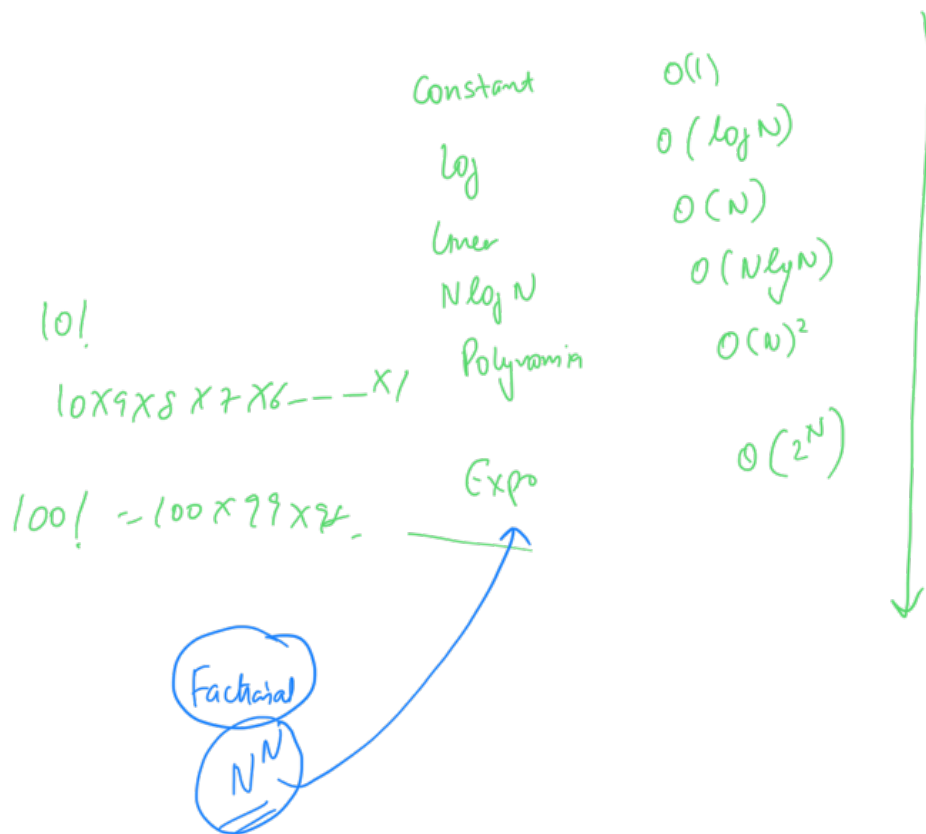
$$2^9 = 512$$

$$2^{15} = 32,768$$

$$2^{20} = 1,048,576$$

$$2^{30} = 1,073,741,824$$

$$2^{40} = 10,995,116,277,76$$



⑧ Find out where will factorial lie in the above landscape!!

$N! \neq O(N)$ ☹️ ☹️
☹️

$100!$
6

$100 \times 99 \times 98 \times 97 \times 96$

10:58p

11:02p

2^N

N^N

$O(N^2) =$

$f(n) =$

$10^5 + 10^{12} + N^2 + N \log$

$O(N^2)$

① $O(3^N) = f(n) = 3^N + 3^{1000} + N^2$

Arrows point from 3^N to $O(3^N)$, from 3^{1000} to $O(1)$, and from N^2 to $O(N^2)$.

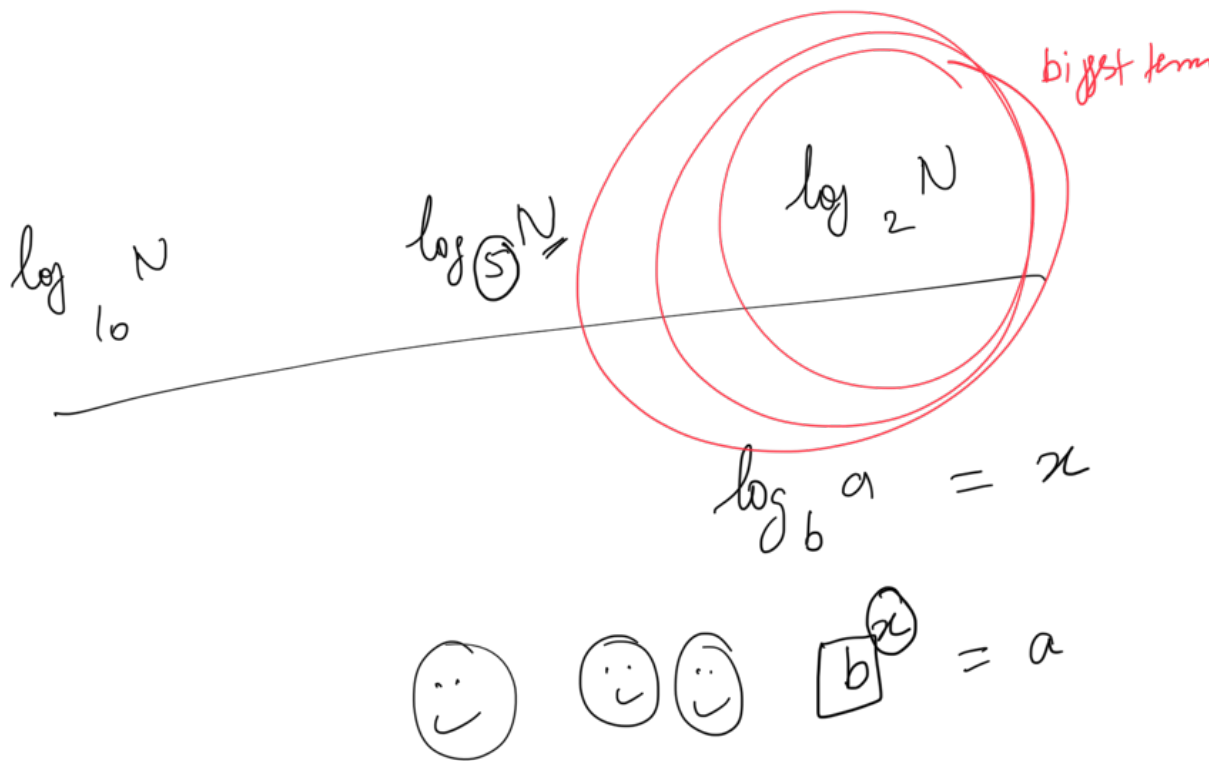
② $O(3^N) = f(n) = 2^5 + 2^N + 3 \times 2^N + 3^N$

Arrows point from 2^5 to $O(1)$, from 2^N to $O(2^N)$, from 3×2^N to $O(2^N)$, and from 3^N to $O(3^N)$.

$4N + 3N \log N + 10^6$

Arrows point from $4N$ to $O(N)$, from $3N \log N$ to $O(N \log N)$, and from 10^6 to $O(1)$.

$O(N)$



Smaller base $\rightarrow$ bigger power

$$N = 10$$

$$\log_{10} 10 = 1$$

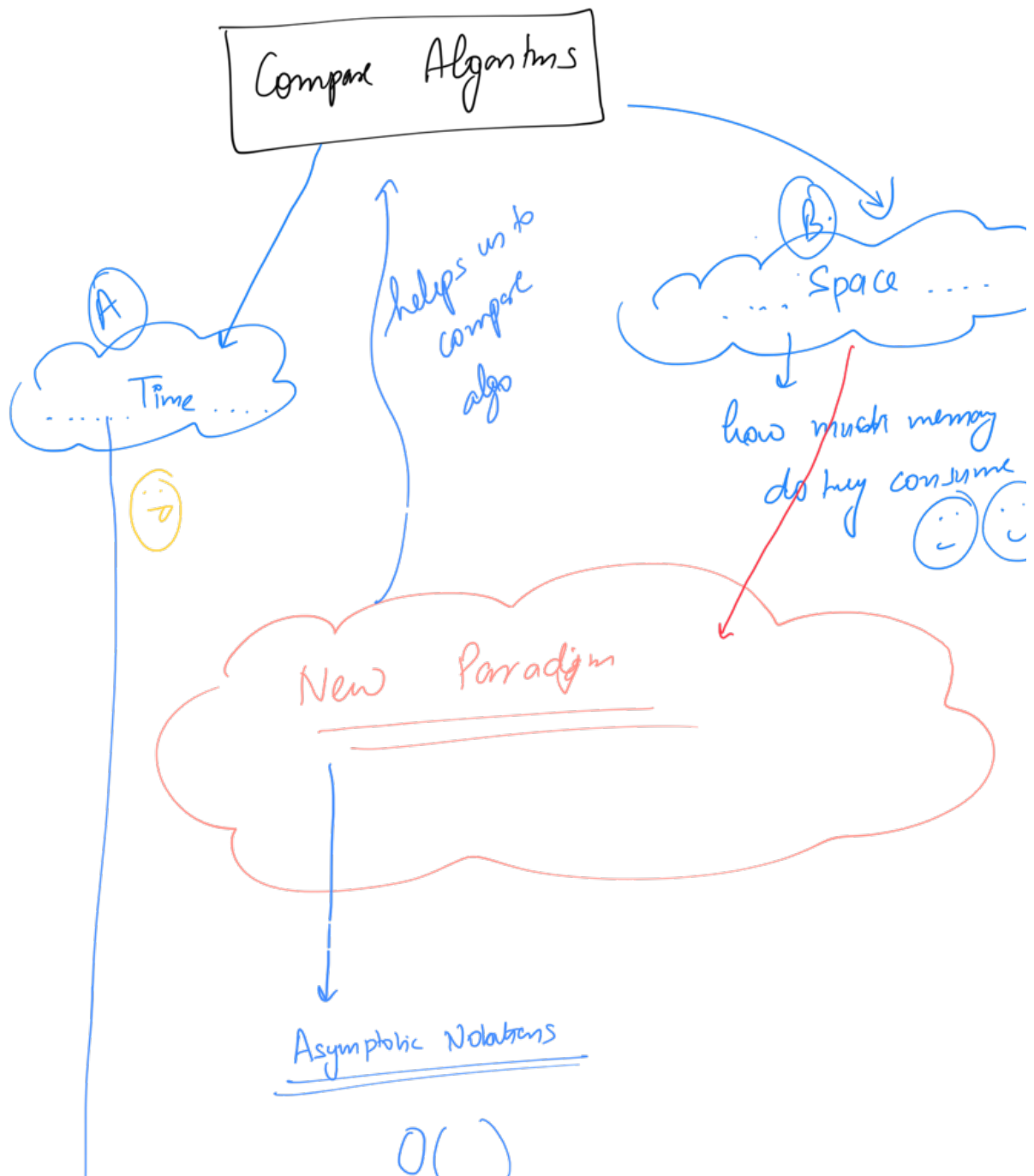
$$\boxed{\log_2 10} = 3..$$

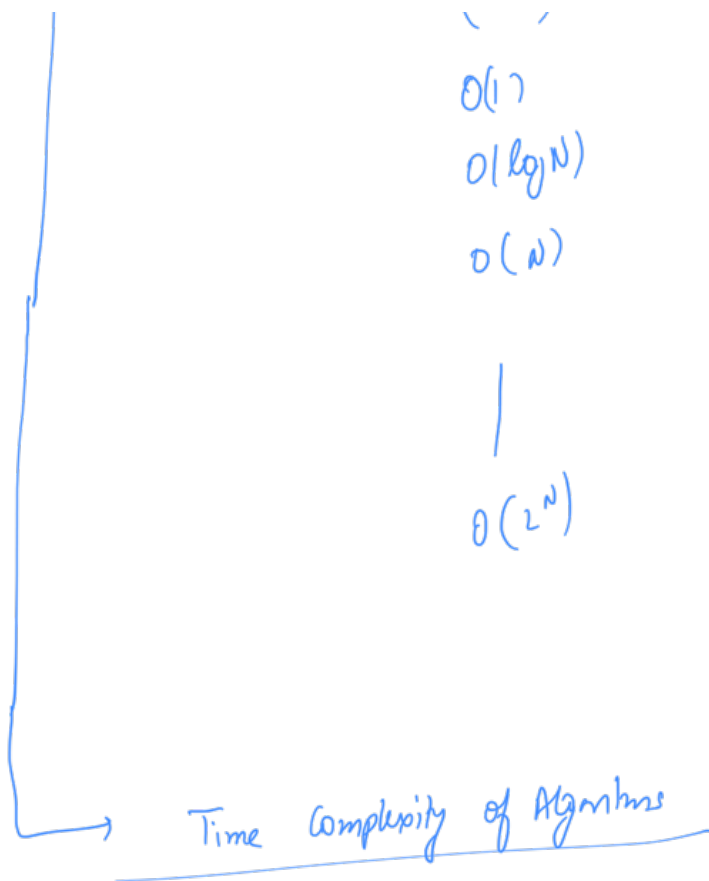
$$\log_{10} 100 = \underline{\underline{2}}$$

$$\boxed{\log_2 100} = (6.7)$$



Time Complexity





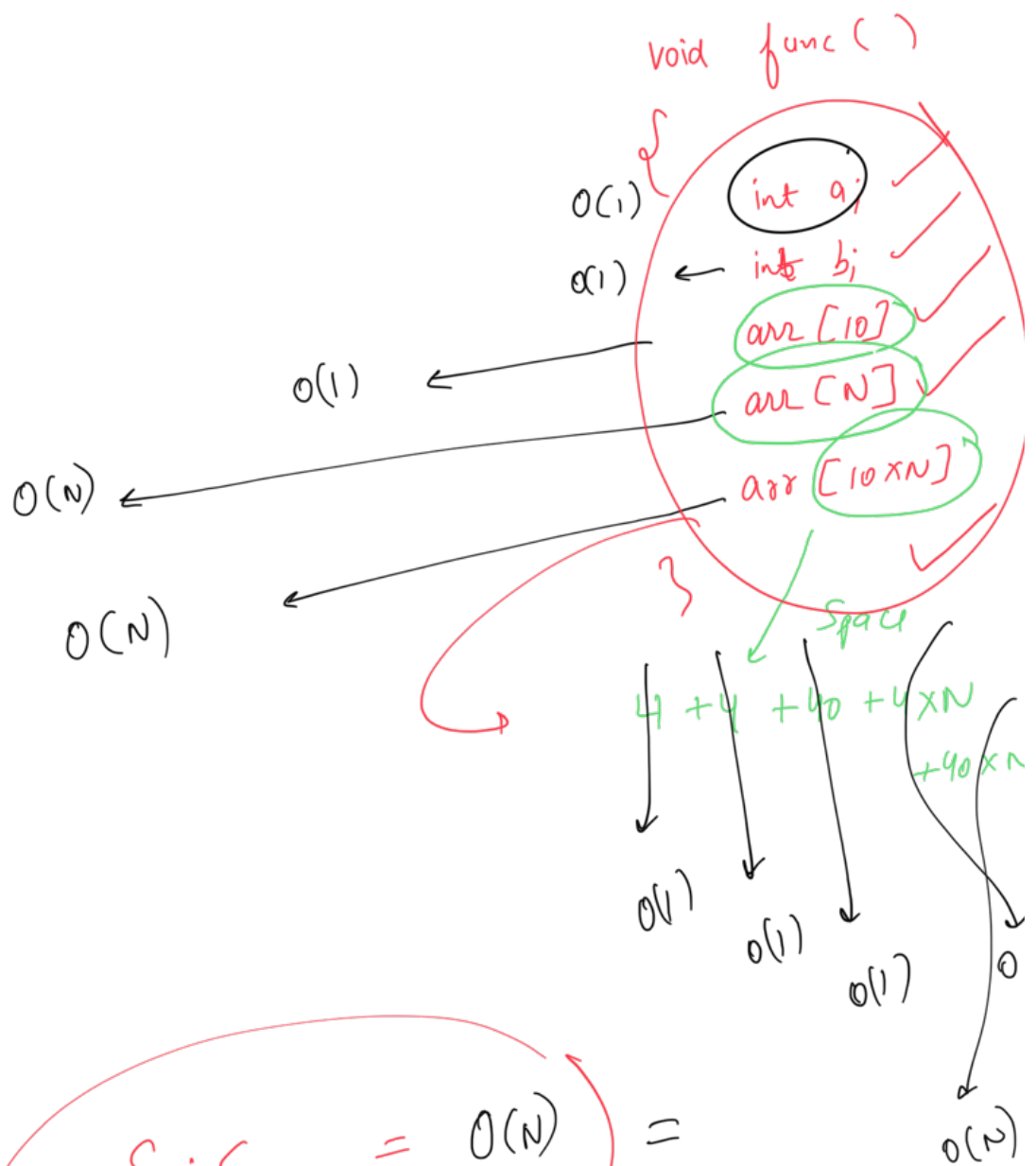
Algo 1 : # computation or # iterations $O(N^2) = 4N^2 + 10N$	Algo 2: $O(N^3) = 10N^3 + 100N$
---	--

Algo 1 runs faster than Algo 2
(asymptotically)

T.C = $O()$

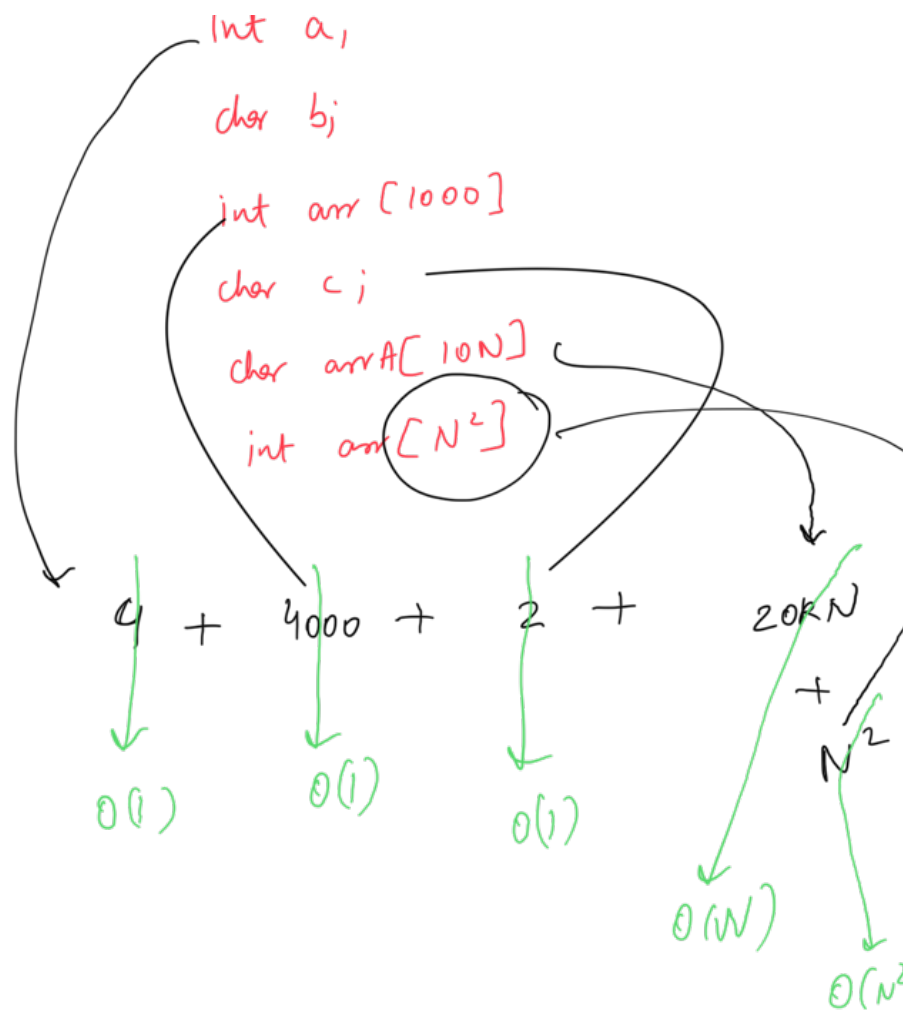
↓

Time Complexity

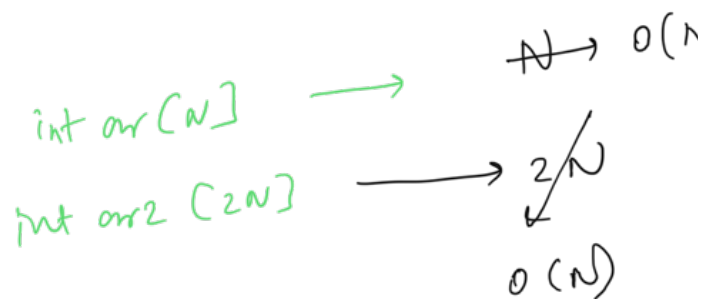


S.C = $O(N)$ =





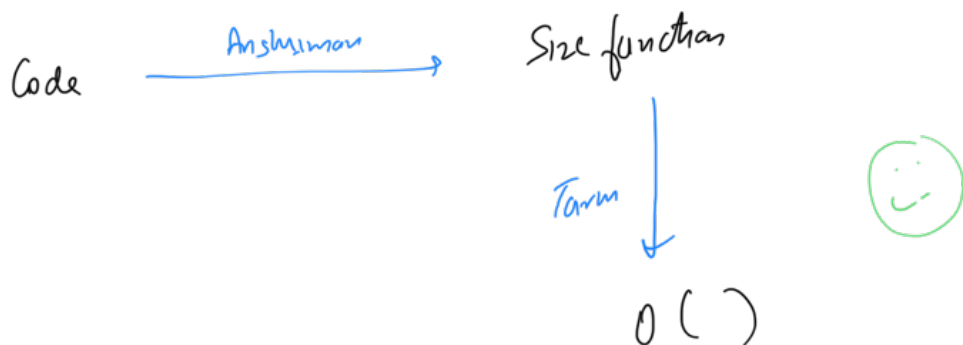
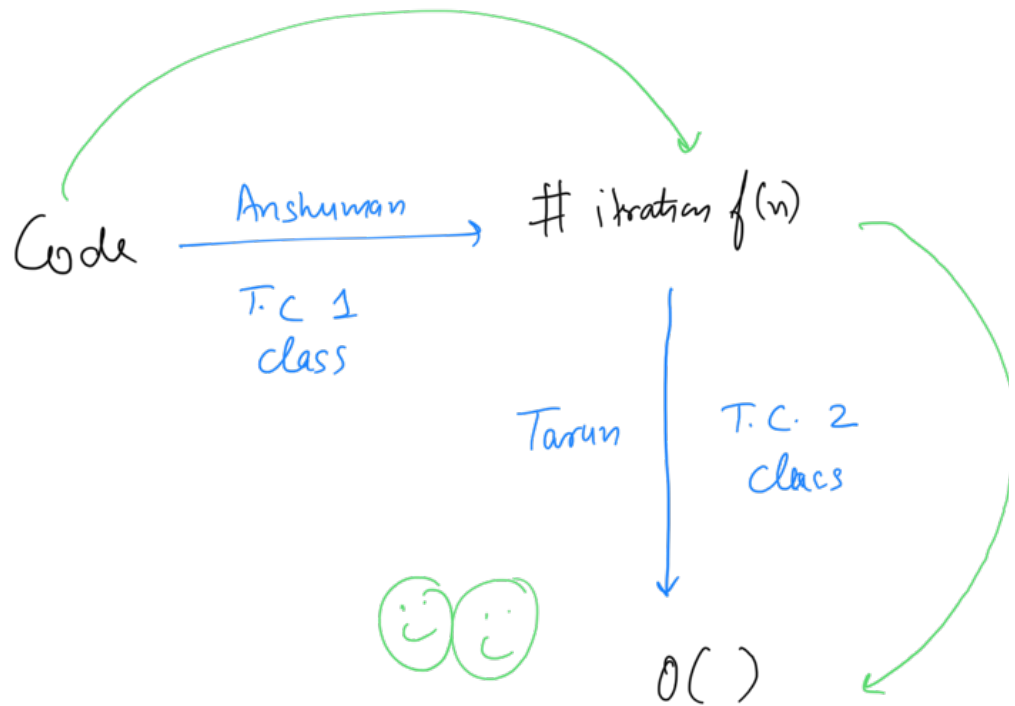
S.C = $O(N^2)$



S.C = $O(N)$



$$T.C = O(1)$$
$$S.C = O(1)$$



Assume every data type to be constant space

int func (int arr[N], int N)

iterations f()
 T.C = ~~N~~ → O(N) 😊

```

sum = 0
for ( i = 0, i < N, i++ )
{
    sum = sum + arr[i]
}

return sum;

```

Amount of Space Function

$N + 1 + 1 + 1$
 ↓ ↓ ↓ ↓
 $O(N)$ $O(1)$ $O(1)$ $O(1)$

S.C = O(N) 😊

	next char
Auxillary space	
Input space	😊 (i